



Answers & Solutions:

1. **B. to prevent sky wave and upper ray interference**
2. **D. may occur around the edge of a sharp obstacle**
3. **B. 0.99**

Solution:

$$n = \sqrt{1 - \frac{81n}{f^2}}$$

$$n = \sqrt{1 - \frac{81(100 \times 10^6)}{(5 \times 10^6)^2}}$$

$$n = 0.99 \rightarrow \text{Ans.}$$

4. **A. window**
5. **D. Solar radiation ionizing the outer atmosphere**
6. **A. 30.00 MHz**

Solution:

$$\cos \theta = \frac{f_c}{\text{MUF}}$$

$$\text{MUF} = \frac{f_c}{\cos \theta} = \frac{25 \text{ MHz}}{\cos 45}$$

$$\text{MUF} = 35.35 \text{ MHz}$$

The best frequency to use = OWF

$$\text{OWF} = 0.85 \text{MUF} = (0.85)(35.35)$$

$$\text{OWF} = 30 \text{ MHz}$$

Note:

OWF = Optimum Working Frequency

7. **B. Transverse**
8. **A. 5775 kHz**

Solution:

$$\cos \theta = \frac{f_c}{\text{MUF}}$$

$$f_c = \text{MUF} \cos \theta$$

$$f_c = (7050.50 \text{ kHz}) \cos 35^\circ$$

$$f_c = 5.775 \text{ MHz}$$

$$f_c = 5775 \text{ kHz} \rightarrow \text{Ans.}$$

9. **D. affected by the solar cycle**
10. **B. below the F layer**
11. **D. reflection and refraction from the ionosphere**
12. **B. 26.5 W/m²**

Solution:

$$\phi = \frac{\xi^2}{\alpha} = \frac{(100)^2}{377}$$

$$\phi = 26.5 \text{ W/m}^2 \rightarrow \text{Ans.}$$

13. **A. Radiation**
14. **C. By direct wave**
15. **A. pass into a medium of different dielectric constant**
16. **A. The D region**
17. **C. 2.51**

Solution:

$$\epsilon_r = \left(\frac{\alpha}{Z_o} \right)^2 \left(\frac{1}{\mu_r} \right)$$

$$\epsilon_r = \left(\frac{377}{150} \right)^2 \left(\frac{1}{1} \right)$$

$$\epsilon_r = 2.51 \rightarrow \text{Ans.}$$

18. **C. By direct wave**
19. **B. Fading**
20. **C. 153.313 km**

Solution:

$$\xi_o = \sqrt{30 P_T G}$$

$$\xi_o = \sqrt{(30)(150)(5.17)}$$

$$\xi_o = 152.53 \text{ V}$$

$$\xi_R = \xi_o \left(\frac{4\pi h_T h_R}{\lambda d^2} \right)$$

$$d = \sqrt{\left(\frac{\xi_o}{\xi_R} \right) \left(\frac{4\pi h_T h_R}{\lambda} \right)}$$

$$d = \sqrt{\left(\frac{152.53 \text{ V}}{22 \frac{\mu\text{V}}{\text{m}}} \right) \left(\frac{4\pi (50 \text{ m})(30 \text{ m})}{5.56 \text{ m}} \right)}$$

$$d = 153.313 \text{ km} \rightarrow \text{Ans.}$$

21. **D. tilting**
 22. **C. the distance a wave travels in one period**
 23. **A. 14.14 mi**
- Solution:
- $$d = \sqrt{2h_t} = \sqrt{2(100)}$$
- $$d = 14.14 \text{ mi} \rightarrow \text{Ans.}$$
24. **B. Isotropic point source**
 25. **C. By direct wave**

END